

## Mastering GitHub Actions Matrix Builds Easily

Testing your code on different operating systems and language versions manually takes hours of tedious work. GitHub Actions matrix builds automate this entire process, running your tests everywhere simultaneously with just a few lines of code.

### What you'll learn:

- 1 Setting Up Your First Multi-OS Matrix
- 2 Testing Multiple Language Versions
- 3 Excluding Unsupported Combinations
- 4 Adding Custom Variables with Include

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# 1

TIP 1 OF 4

## Setting Up Your First Multi-OS Matrix

Instead of writing separate workflows for Windows, Mac, and Linux, you can define a matrix strategy. GitHub will spin up separate virtual machines for each OS you list and run your steps on all of them in parallel. This ensures your code is truly cross-platform without any manual effort.

```
$jobs:
  test:
    runs-on: ${{ matrix.os }}
    strategy:
      matrix:
        os: [ubuntu-latest, windows-latest, macos-latest]
```

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# 2

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## Testing Multiple Language Versions

Users of your software might be running older or newer versions of Node.js, Python, or Go. By adding a version array to your matrix, GitHub Actions will cross-multiply your operating systems with your language versions, testing every possible combination automatically.

```
$strategy:
  matrix:
    os: [ubuntu-latest, windows-latest]
    node: [18, 20]
  steps:
    - uses: actions/setup-node@v4
      with:
        node-version: ${{ matrix.node }}
```

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# 3

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## Excluding Unsupported Combinations

Sometimes, certain combinations simply don't make sense or are known to fail (like running an old legacy version of Python on the newest macOS). You can use the 'exclude' keyword to tell GitHub to skip these specific runs, saving you precious build minutes and avoiding false alarms.

```
$strategy:
  matrix:
    os: [ubuntu-latest, windows-latest, macos-latest]
    node: [18, 20]
    exclude:
      - os: macos-latest
        node: 18
```

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## 4

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### Adding Custom Variables with Include

If you want to inject a specific environment variable or tool version into just one specific combination in your matrix, you can use the 'include' keyword. This lets you add extra configuration values to a run without having to expand the entire matrix for everyone else.

```
$strategy:
  matrix:
    os: [ubuntu-latest, windows-latest]
    include:
      - os: ubuntu-latest
        experimental: true
        custom_flag: "--fast"
```

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## Key Takeaways

- + Matrix builds run jobs in parallel by default, saving you time.
- + Use `${{ matrix.variable_name }}` to reference matrix values in your steps.
- + GitHub charges by build minutes, so keep your matrices reasonably sized.
- + The total number of jobs is calculated by multiplying all matrix arrays together.
- + Use 'fail-fast: false' if you don't want one failing OS to cancel all other running tests.

## Watch Out For

- ! Creating giant matrices (e.g., 3 OS x 4 Versions = 12 jobs) on every single git push, which quickly drains your free action minutes.
- ! Forgetting that file paths are written differently on Windows (backslash) vs Linux/macOS (forward slash) in shell scripts.
- ! Not setting 'fail-fast: false', causing a minor Windows-specific bug to immediately stop your successful Linux and Mac builds from finishing.

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